

Project – Controlled landing on moving platform

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ABSTRACT

The objective of this project is to make the Parrot Mambo Drone to hover over a moving landing platform

1. INTRODUCTION

This lab focuses on developing a Simulink-based model that enables the Parrot Mini drone to hover along a line following robot on top of the line following robot there is a landing pad on which the drone should land.

2. METHOD

The Parrot Mambo drone is connected to the host computer via Bluetooth, ensure the connection by pinging the IP “192.168.3.1”, the MATLAB application is opened and the **parrotMinidroneKeyboardControl** command is prompted and the Simulink instance is started.

By opening the Flight control system model and going to then to the path planning sub model. We can observe that we already vision-based data and the Landing enable. These are the sub models which we are going to work with. We are going to introduce a state flow chart which will handle the drone movement.

2.1 Position Control Logic Override

From the stateflow chart, the outputs xestimate, yestimate, yawestimate, x_command, y_command and targetdetectflag are then passed to the state flow chart as inputs. The state flow chart's outputs are xout, yout, zout, yawout, visionland and controlflag. Here I have deleted the functionality of yawout.

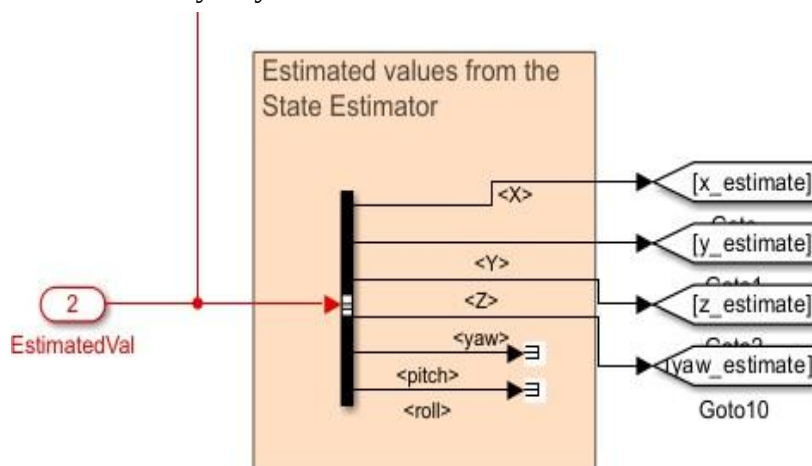


Fig 2.1 State estimator

2.3 Landing Enable

Fig 2.3 Landing Enable Logic

2.4 State Flowchart for Drone movement

The state flowchart illustrates the sequential decision-making cycle whereby Parrot Mambo drone is able to move along a line independently, colored, and land when they see a disc-shaped marker. Takeoff state will make the drone take off at the height of 1.1 m, MoveForward state will make the drone move forward and finally the Land state will trigger the land flag that initiates the land process for the drone.

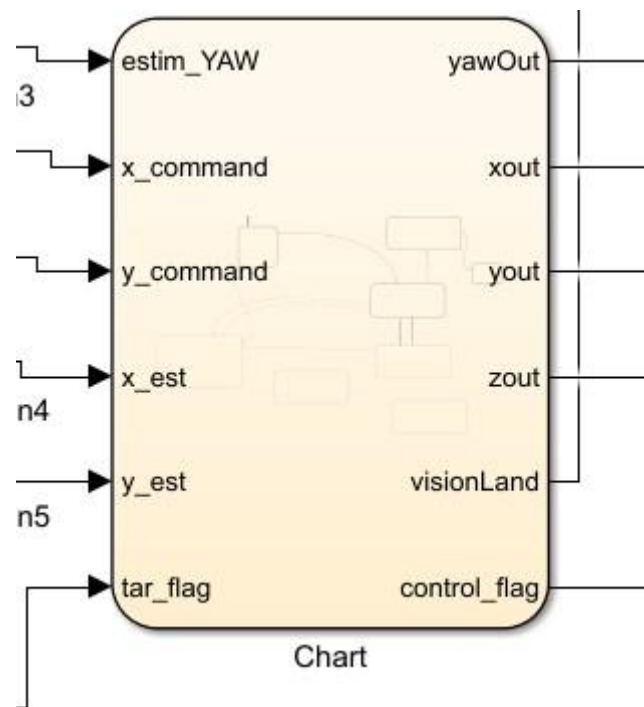


Fig 2.4 State Flow chart overlay

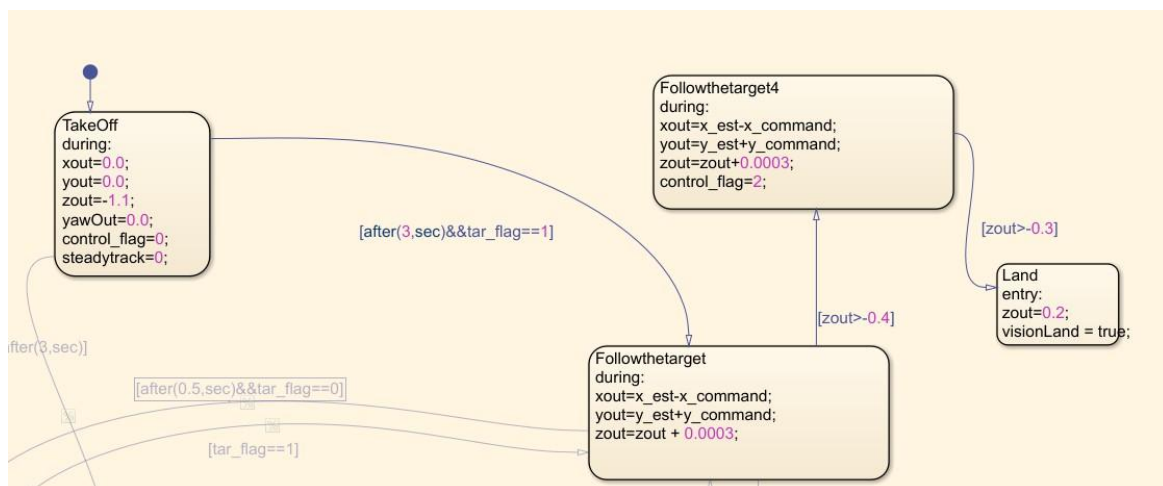


Fig 2.5 State Flow chart Logic

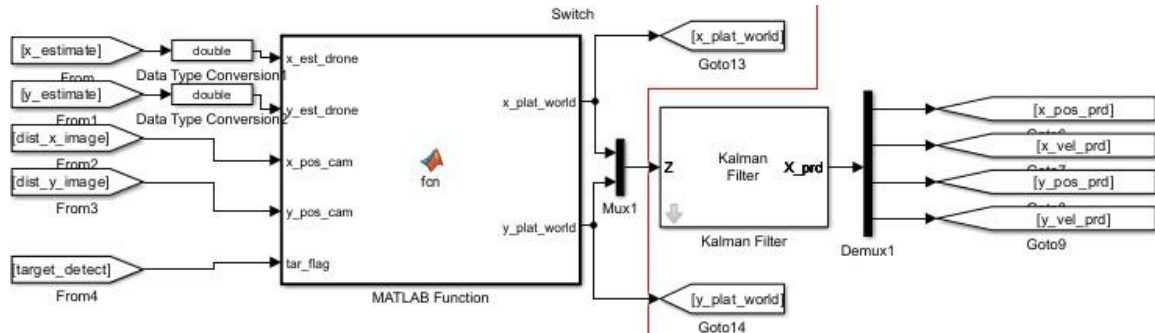


Fig 2.6 X,Y Position prediction to follow the line following robot

3. RESULTS

It is observed that as the program is downloaded to the drone the flight control system will control the drone to the landing pad on the top of the line following robot. And when detected, the drone will slowly descend and once it has reached a certain height it allows the land flag resulting in the end of the flight.